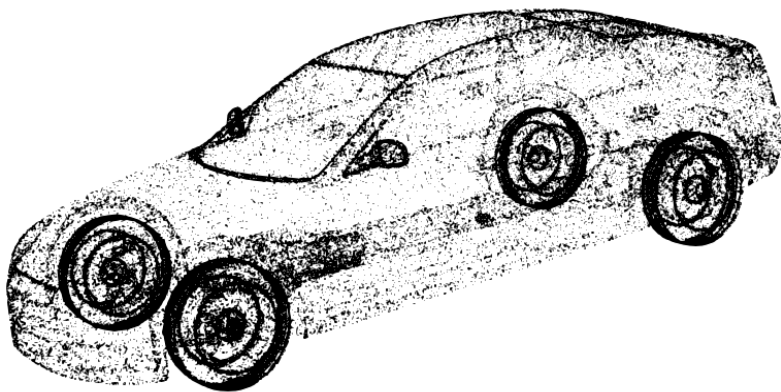
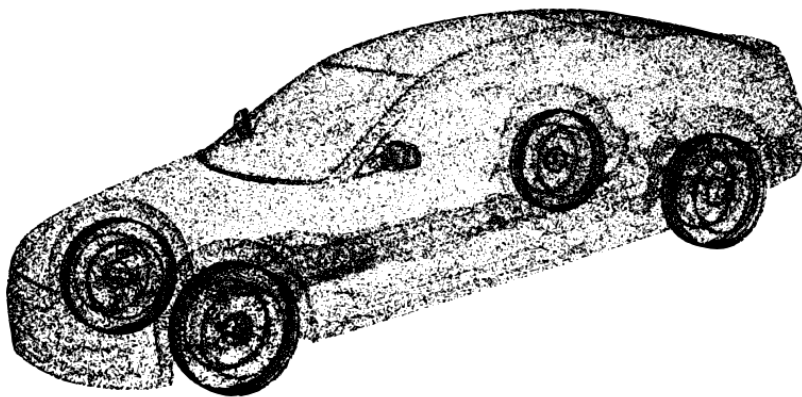


Meeting 01.07.2026

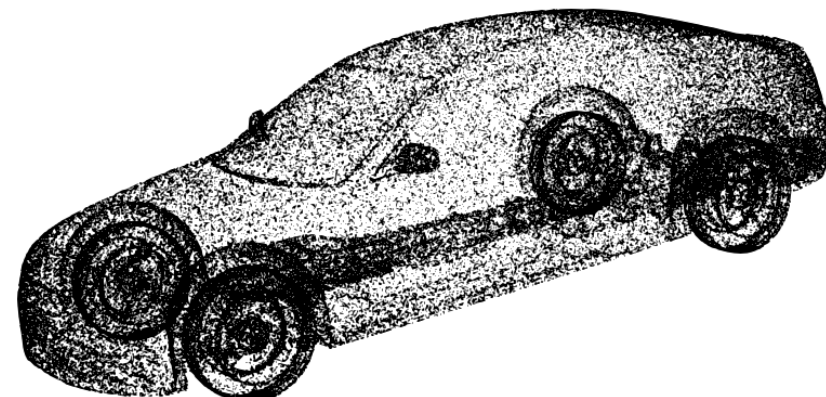
Update with Parsa



Training distribution

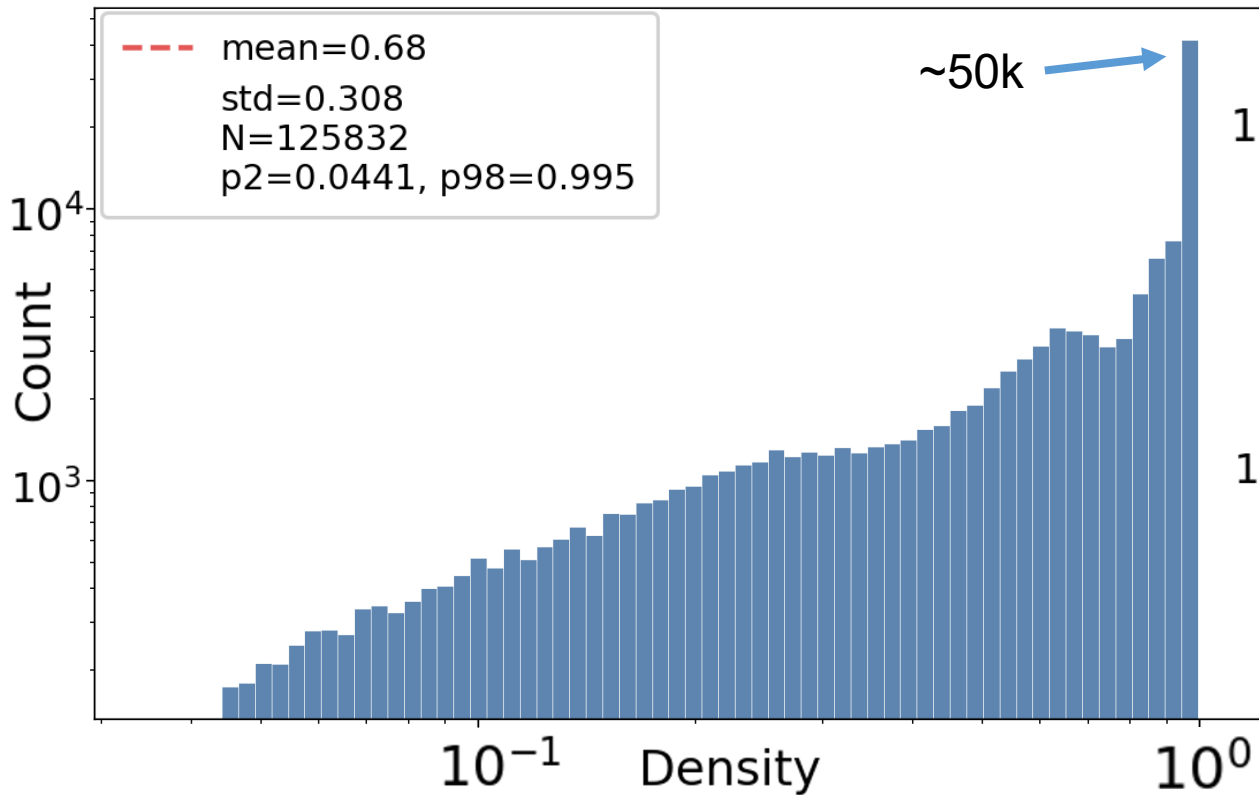


Density correction intensity $\beta=0.5$

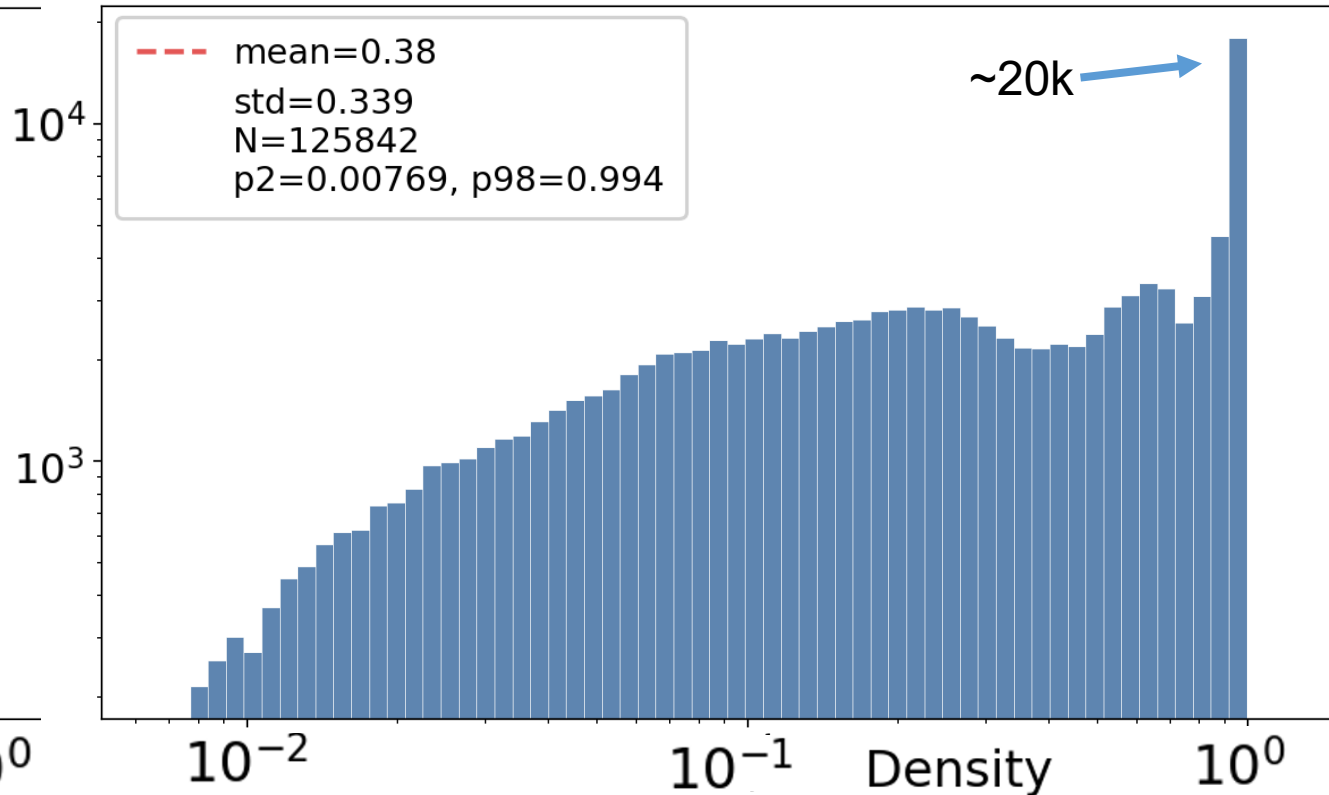


Density correction intensity $\beta=1.0$

Run 34 sampled input density histogram (beta=0.00, points=131072)



Run 34 sampled input density histogram (beta=1.00, points=131072)



Training - Loss Formulation

- Previously we trained on $\beta=0.0$ and $\beta=1.0$ and then tested on β in between 0.0 and 1.0 (**interpolation test – SATLOSS3**)
- Now we train on $\beta=0.0$ and per-batch-random $\beta=[0.1 \text{ to } 0.5]$, then test on β greater than 0.5 (**extrapolation test – SATLOSS5**)
- Also, the base (GMP) is trained on double the number of epochs to make the comparison fair

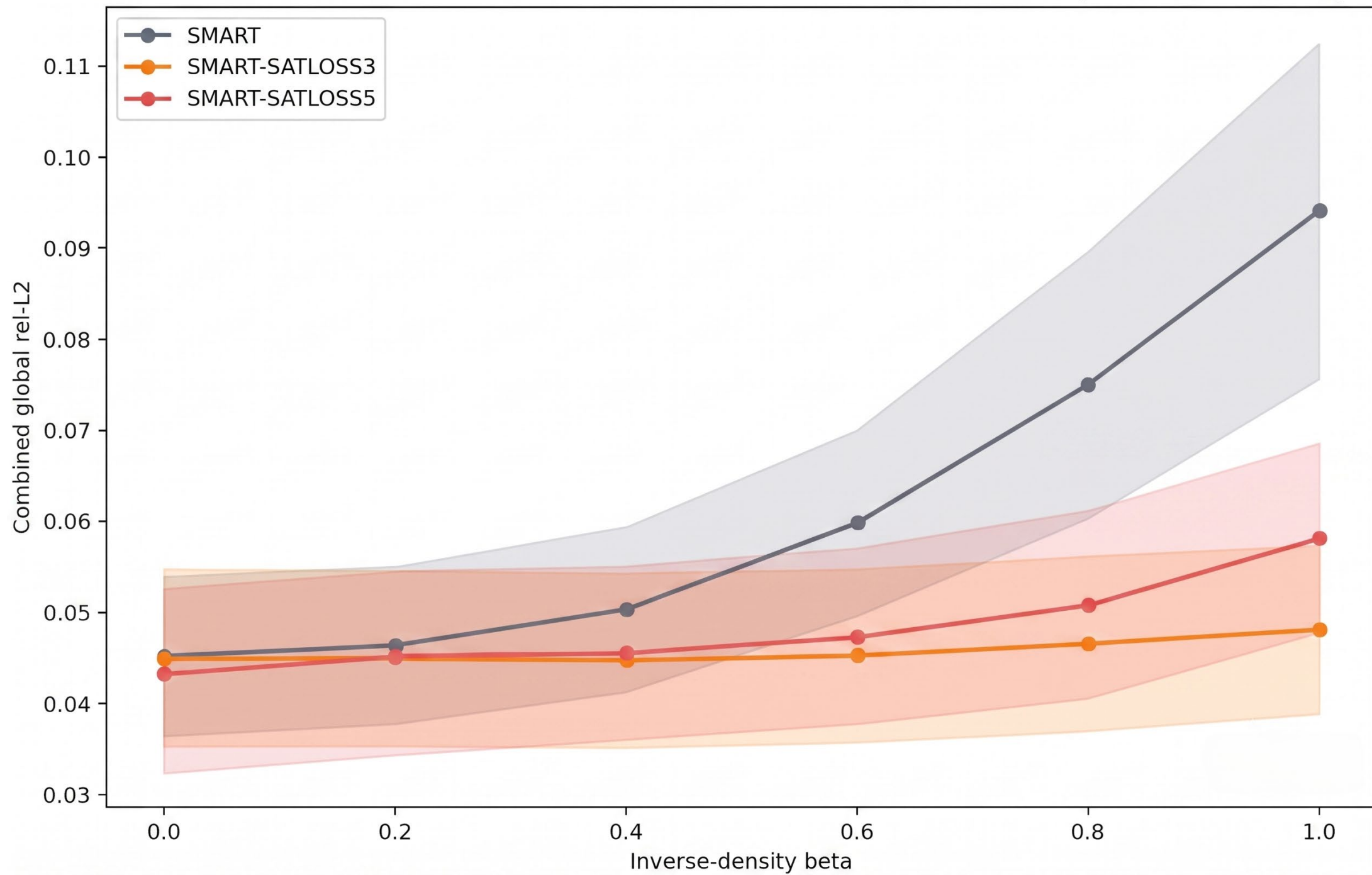
$$L_{\text{primary}} = L(\widehat{y_{\beta=0}}, y_{\text{gt}})$$

$$L_{\text{secondary}} = L(y_{\beta=\text{randomly from } 0.1 \text{ to } 0.5}, \widehat{y_{\text{gt}}})$$

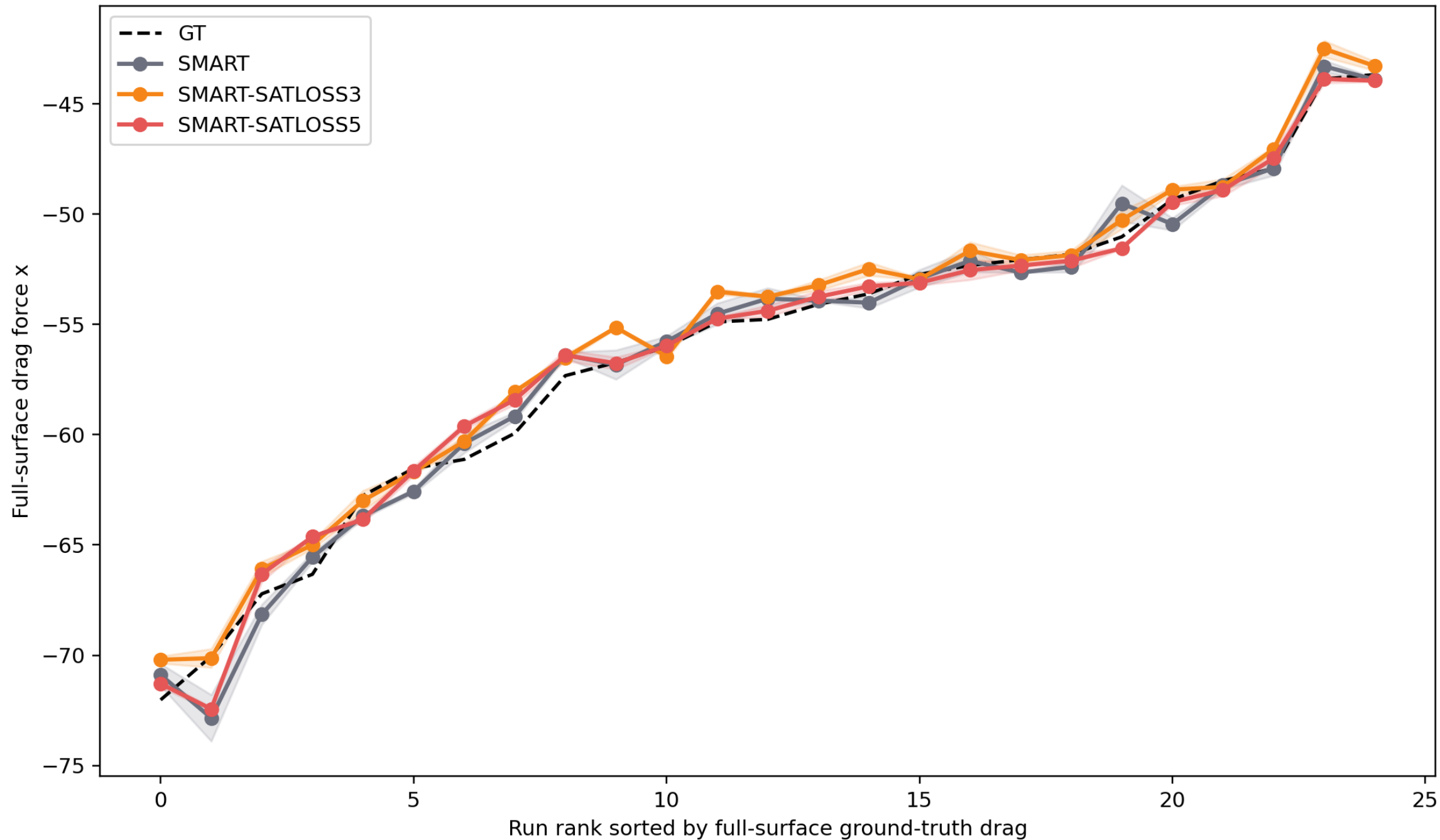
$$L_{\text{pred-match}} = L(\widehat{y_{\beta=0}}, \widehat{y_{\beta=\text{randomly from } 0.1 \text{ to } 0.5}})$$

$$L_{\text{total}} = \text{Sum}(L_{\text{primary}}, L_{\text{secondary}}, L_{\text{pred-match}})$$

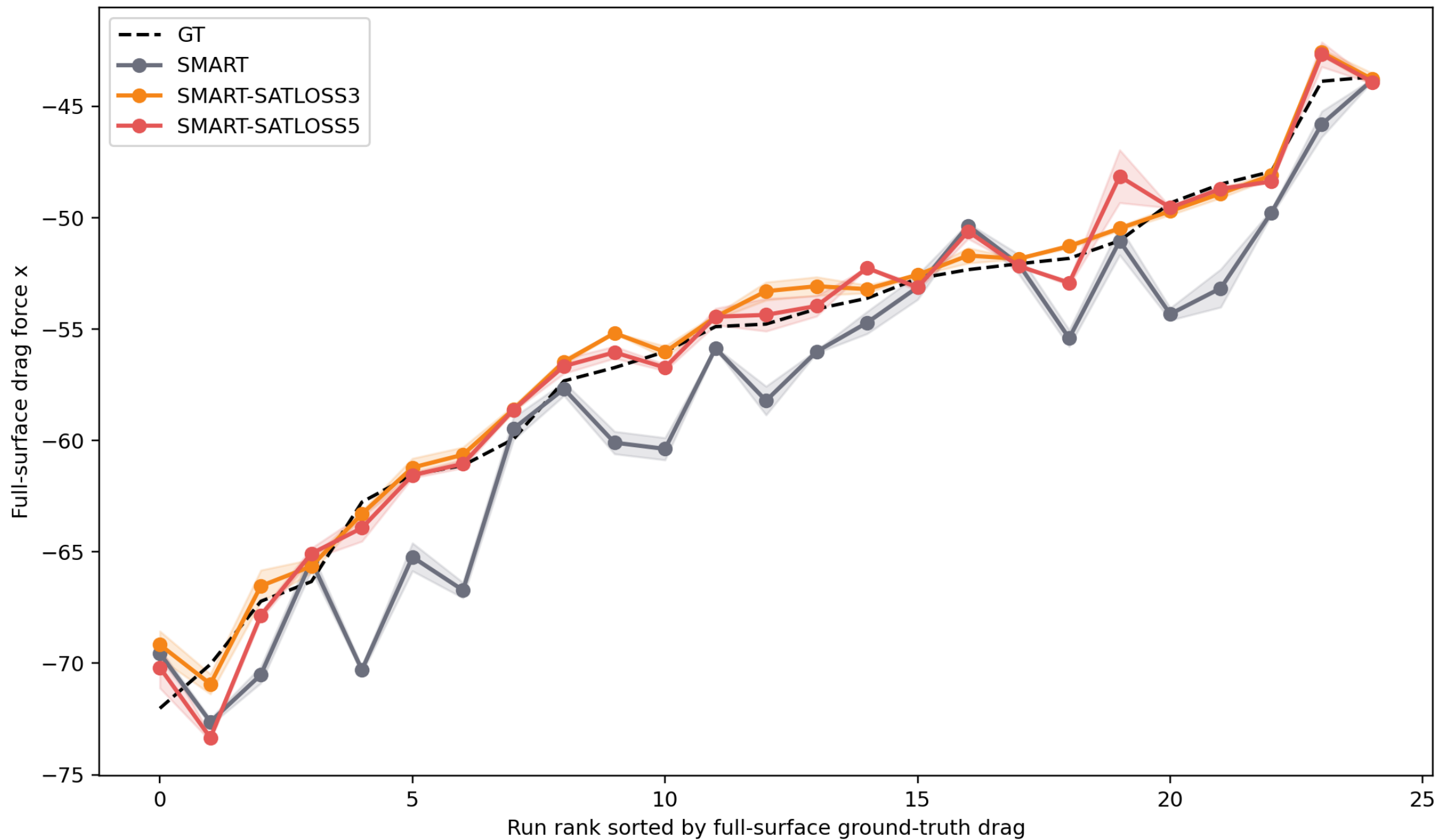
SMART vs SMART-SATLOSS3 vs SMART-SATLOSS5:inverse-density beta severity curve (combined glob

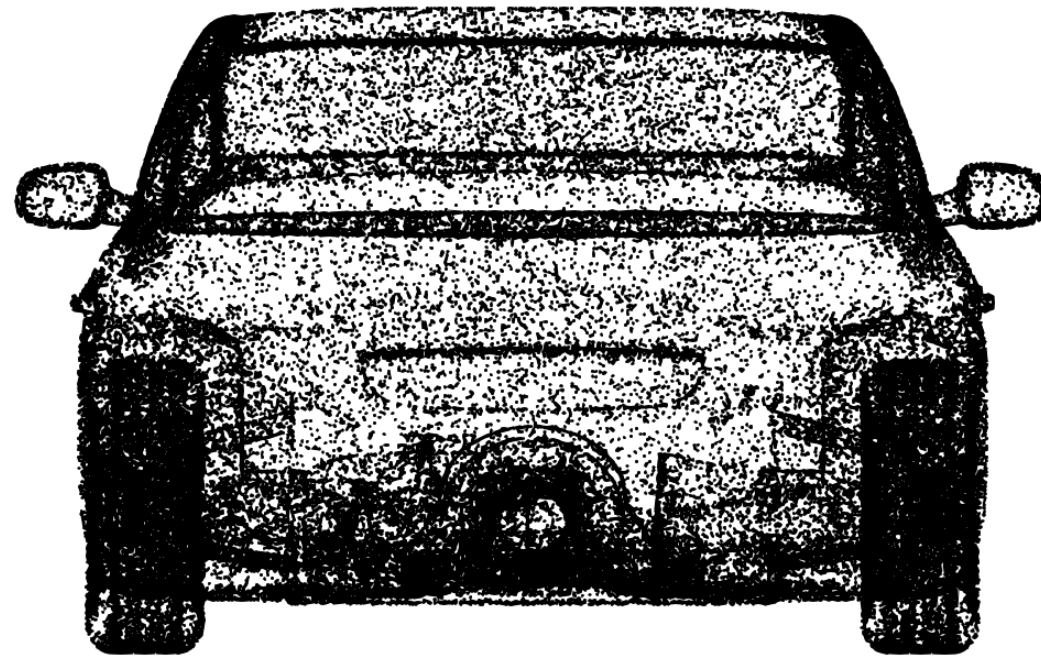
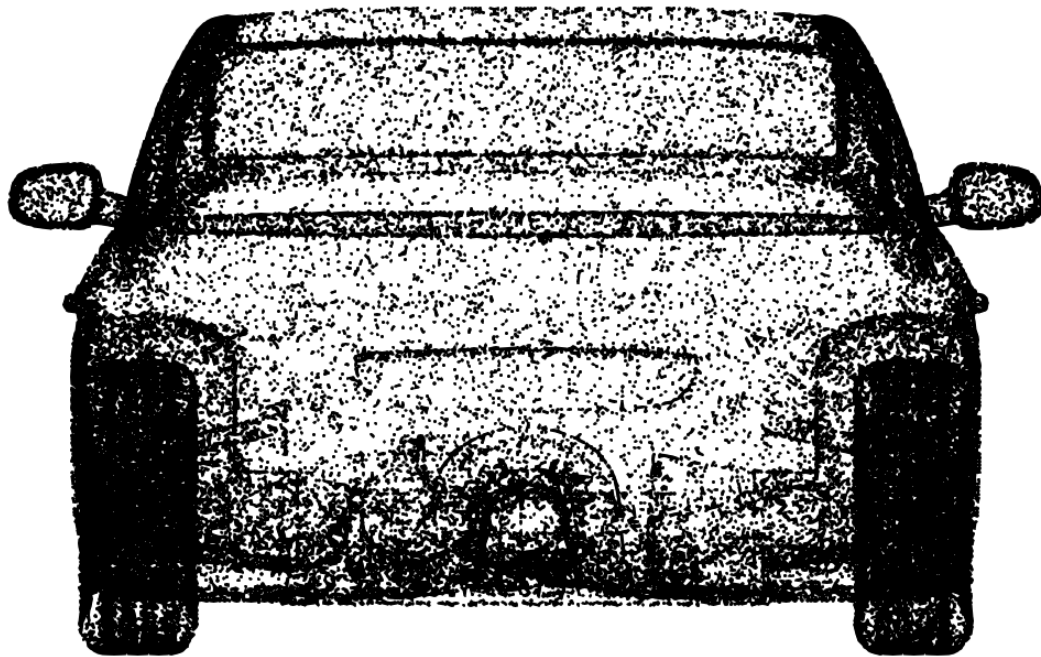


SMART family: full-surface drag ranked by GT drag (beta=0.00)

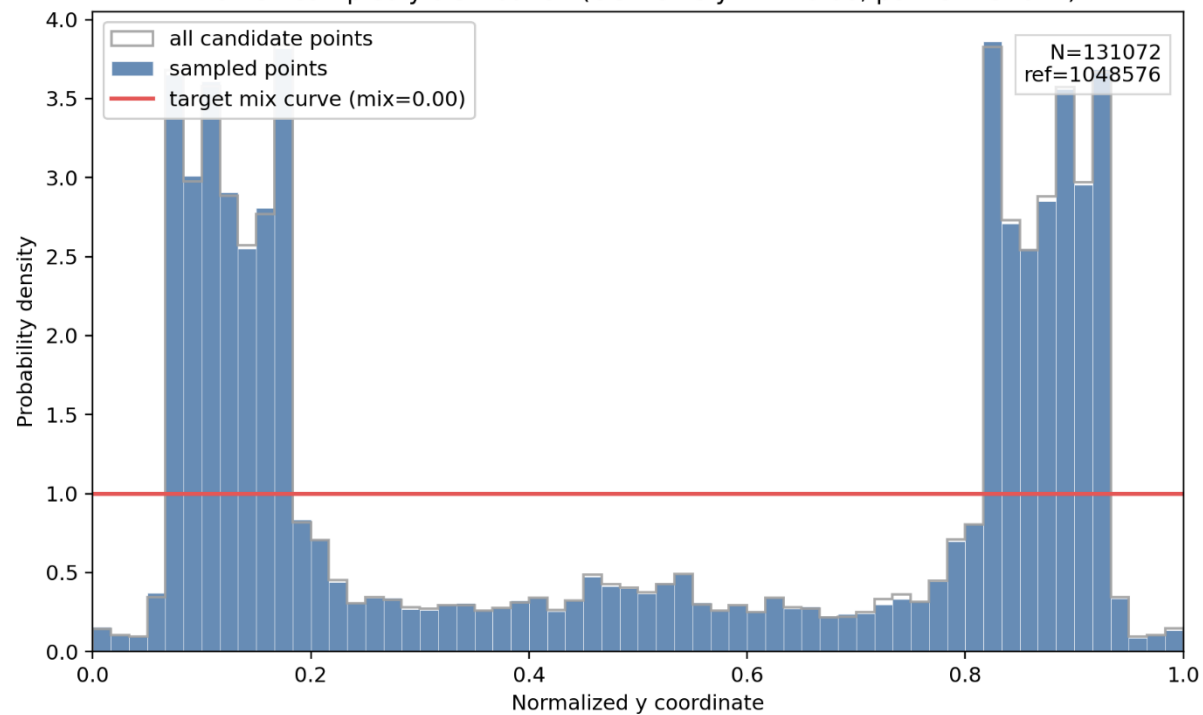


SMART family: full-surface drag ranked by GT drag (beta=1.00)

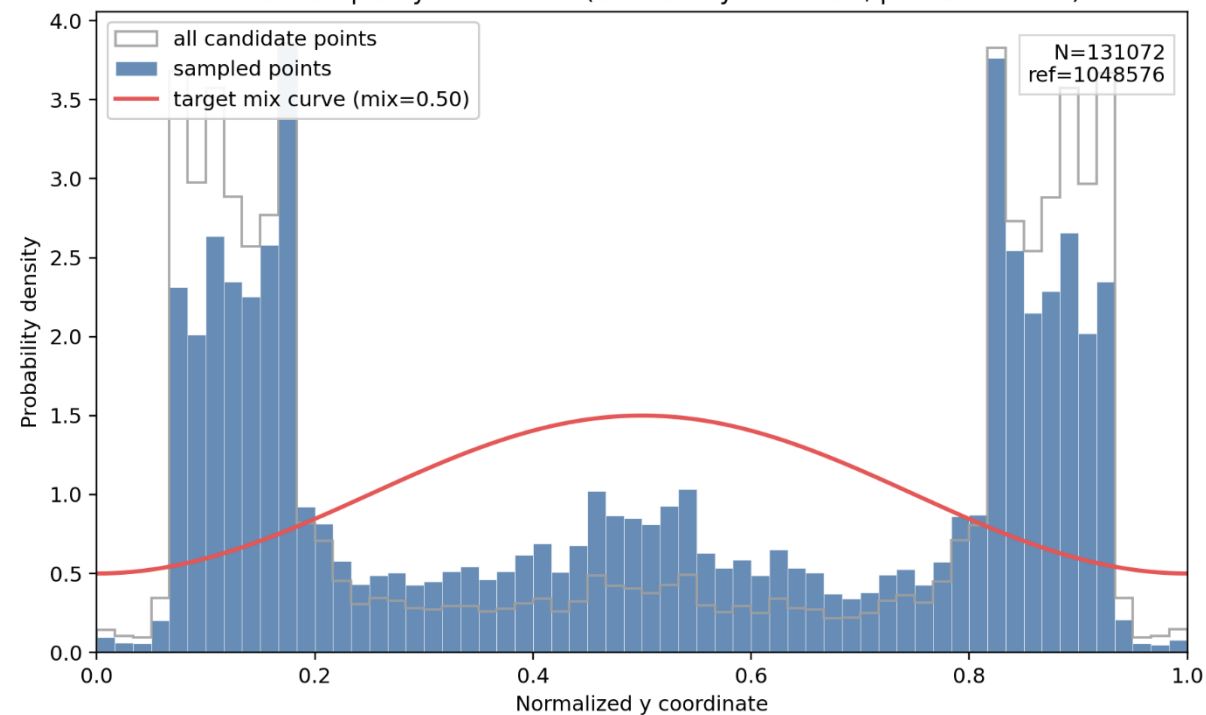




Run 34 sampled y distribution (OOD sine-y mix=0.00, points=131072)



Run 34 sampled y distribution (OOD sine-y mix=0.50, points=131072)



SMART vs SMART-SATLOSS3 vs SMART-SATLOSS5: sinusoidal-y severity curve (combined global)

